



RAI 811: ROBOT MECHANICS & CONTROL



Spatial Descriptions



Spatial Descriptions

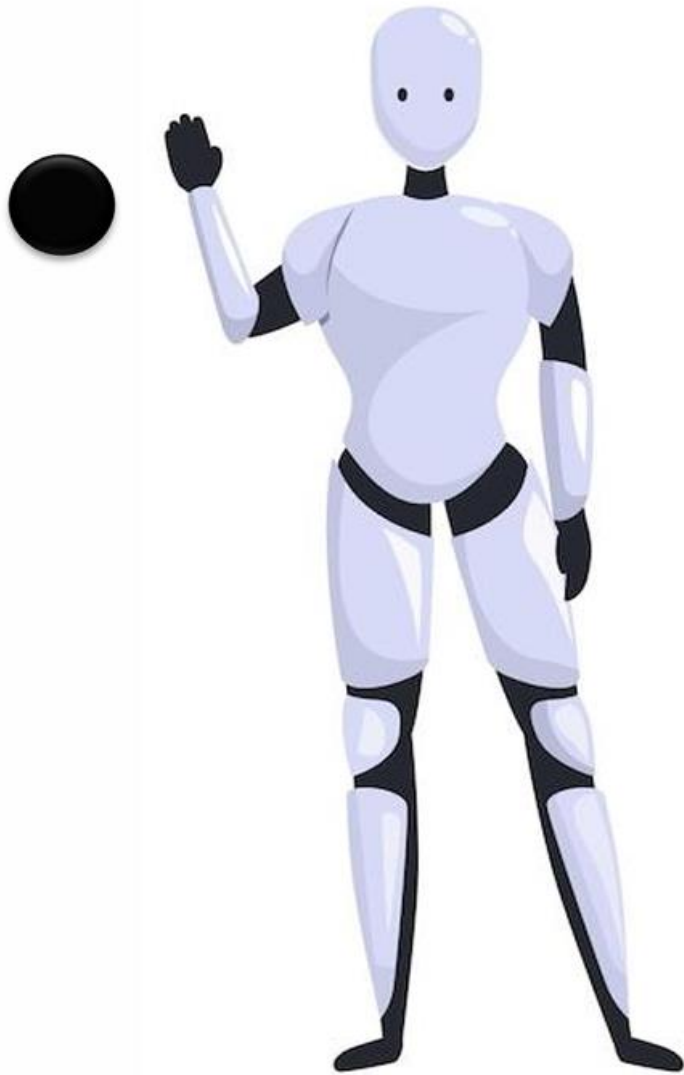
Spatial Descriptions

- Position and Orientation Descriptions





Spatial Descriptions





Point and Vector

- A **vector** is a movement that you can make. like “2 steps north, 3 steps east”.
- A **point** is a position. it’s a place in the room. like “the south east corner” or “the center”.
- Notice that a vector doesn’t say where you start or end. Just how you should move.
- Some operations mix the two. You can take a point and add a vector (start in the center of the room, go 2 steps north and 3 steps east). You can take two points and talk about the vector between them (how do you need to talk to get between the points).



Point and Vector

- Vectors don't normally represent a position, rather they represent a direction and magnitude.
- So, if you subtract a point from another point, you get a vector- subtraction gives you a displacement vector.
- If you add a point and a vector, you get another point.
- **A point is just a vector with a known origin.**
- You can add 2 vectors and get a vector – which is called the resultant vector.



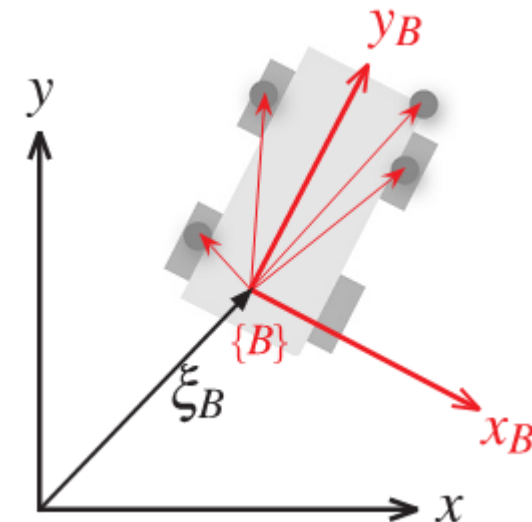
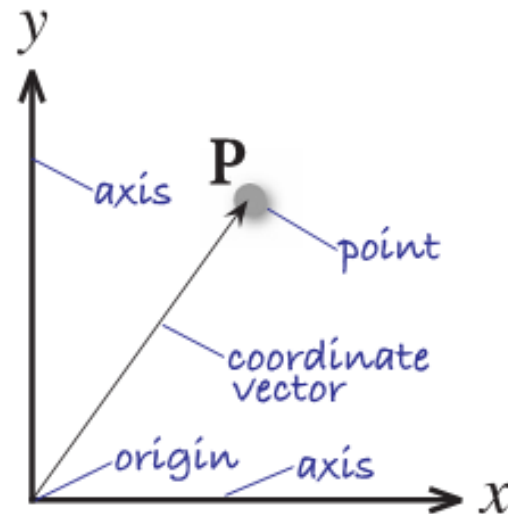
Representing Position and Orientation

- How to describe a position in space?
- How to describe an orientation in space?
- **Pose:** Combination of position and orientation



Representing Position and Orientation

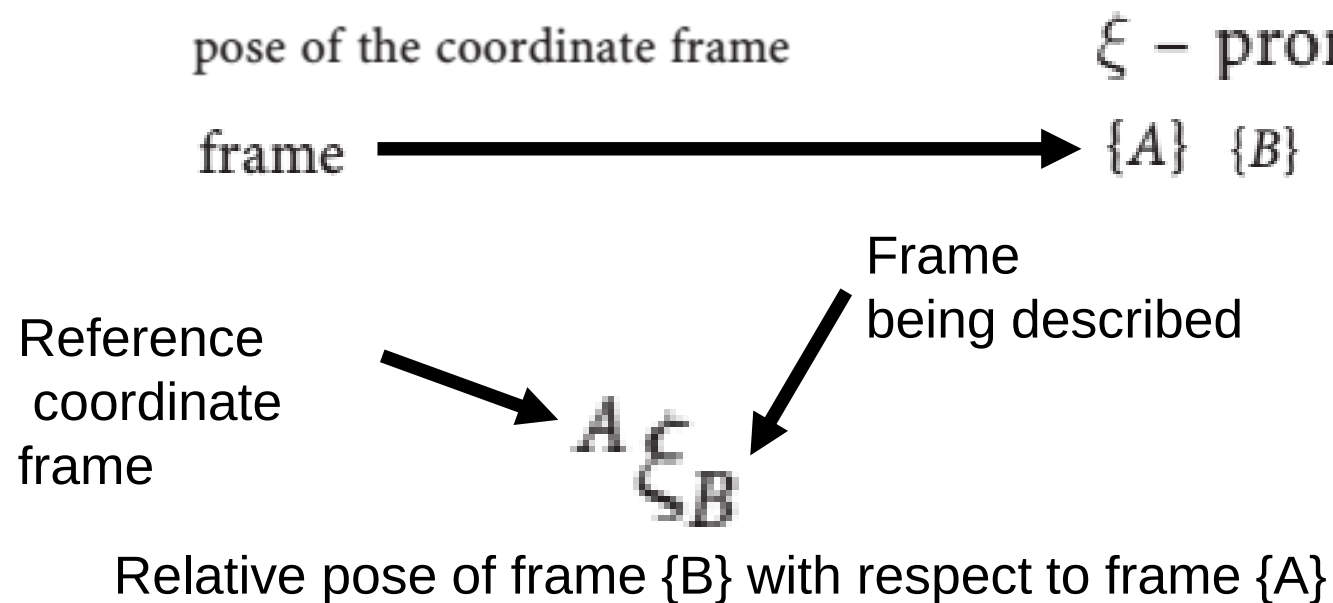
- **Convention:** Attach coordinate frame to an object. It enables us to describe the pose (position and orientation) of an object with respect to reference coordinate frame. Each frame is given a name.
- **Assumption:** object is rigid



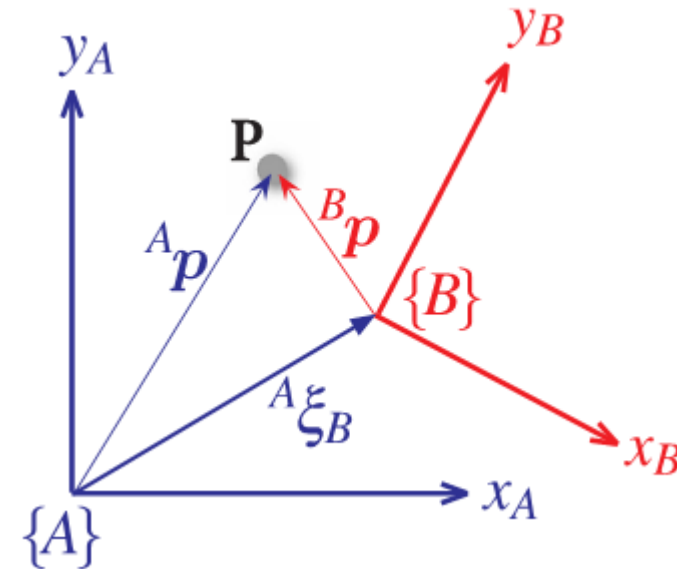


Representing Position and Orientation

- How many dimensions are required to completely describe the pose of an object?



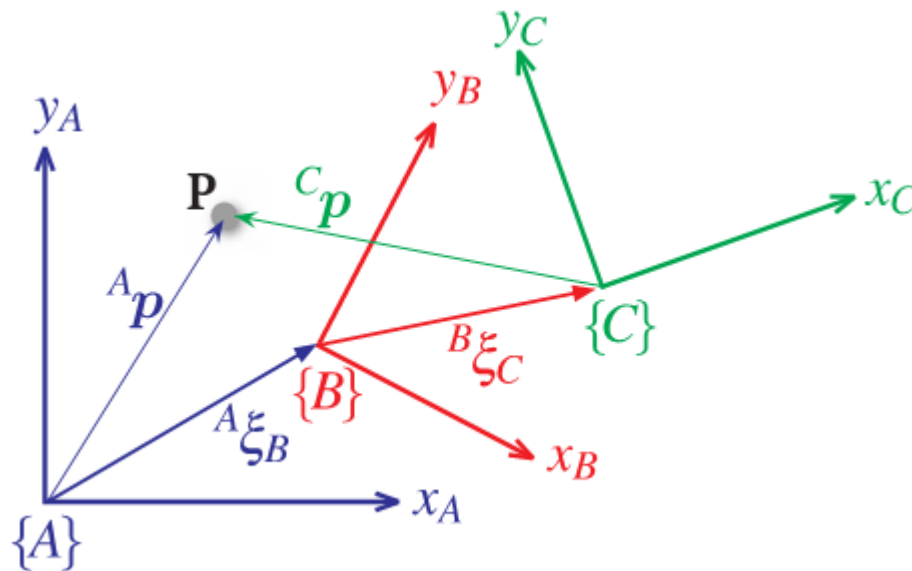
NOTE: If superscript is missing, we assume the change is w.r.t world coordinate frame denoted as $\{O\}$.





Representing Position and Orientation

- An important characteristic of relative poses is that they can be combined/composed together.



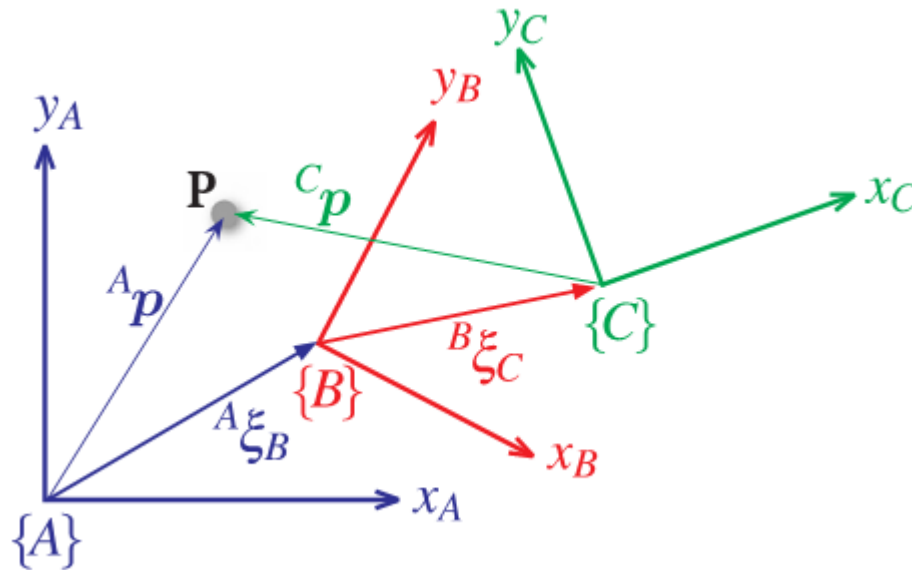
$${}^A\xi_C = {}^A\xi_B \oplus {}^B\xi_C$$

*Pose of {C} relative to {A} can be obtained by compounding the relative poses from {A} to {B} and {B} to {C}. We use the operator $\oplus$ to indicate Composition of relative poses.



Representing Position and Orientation

- An important characteristic of relative poses is that they can be combined/composed together



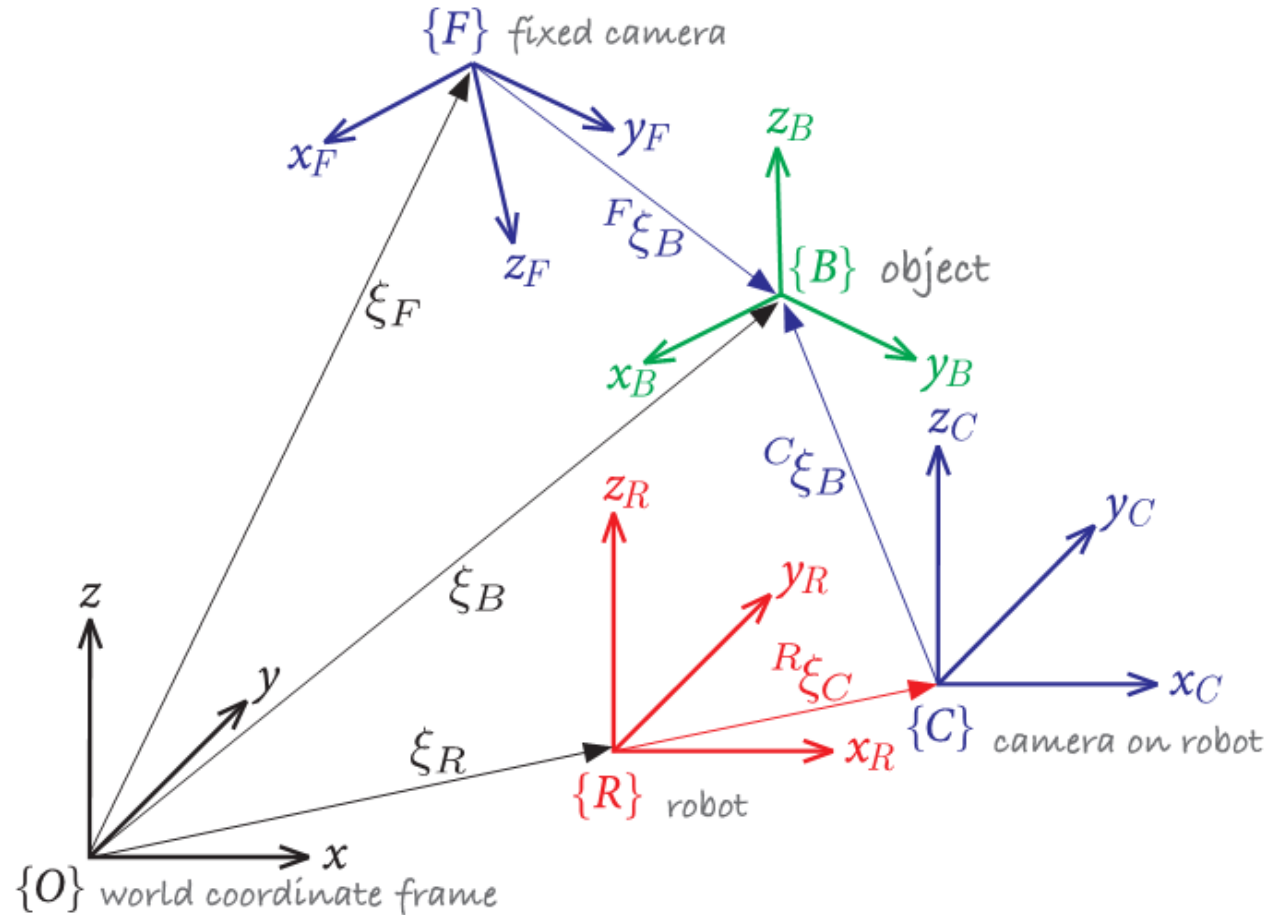
For this case the point $\mathbf{P}$ can be described by

$${}^A \mathbf{p} = \left({}^A \xi_B \oplus {}^B \xi_C \right) \cdot {}^C \mathbf{p}$$

Till now we have seen 2D coordinate frames. It is applicable to many robotic systems especially mobile robots that operate in planar world. However for many other robots, we require 3D coordinate frames.



Representing Position and Orientation



Spatial Relationships

$$\xi_F \oplus {}^F\xi_B = \xi_R \oplus {}^R\xi_C \oplus {}^C\xi_B$$

$$\xi_F \oplus {}^F\xi_R = \xi_R$$



Representing Position and Orientation

- Algebraic Rules:

$$\begin{aligned}\xi \oplus 0 &= \xi, & \xi \ominus 0 &= \xi \\ \xi \ominus \xi &= 0, & \ominus \xi \oplus \xi &= 0\end{aligned}$$

where 0 represents a zero relative pose

$$\ominus {}^X\xi_Y = {}^Y\xi_X$$

A pose has an inverse

$${}^X\xi_Y \oplus {}^Y\xi_Z = {}^X\xi_Z$$

combination

$$\xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1$$

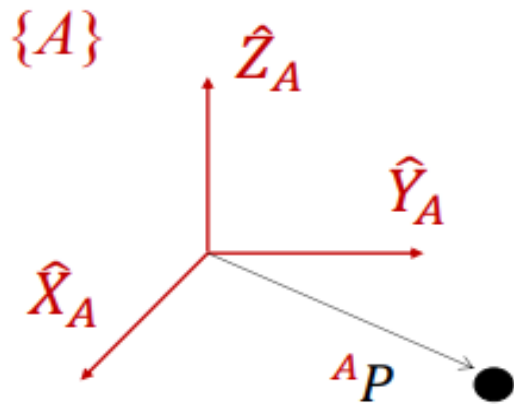
Not commutative with exception
of their sum being zero



Representing Position and Orientation

Description: Position

- The position of any object (or point) in space can be described using a vector with the x , y , and z coordinates
- This vector is called the **position vector** and has size 3×1



$${}^A P = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$



Representing Position and Orientation

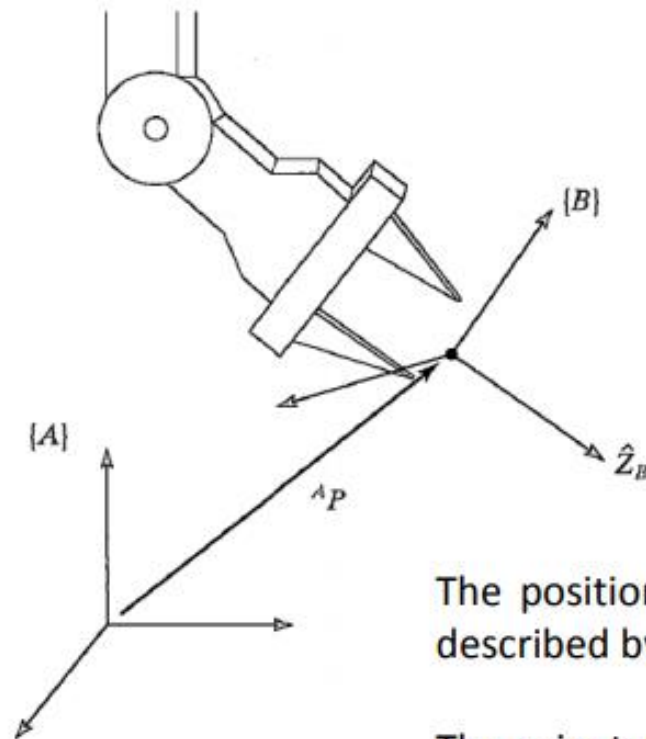
Description: Orientation

- The second property that defines an object in space is its **orientation**
- For example, the end-effector (manipulator) of a robot can have a possible number of orientations at the same point in space
- For describing the orientation of any body in space, a coordinate system is attached to the body
- The coordinate system is then described with reference to another coordinate system



Representing Position and Orientation

Description: Orientation



The position of the end-effector is described by the vector ${}^A P$

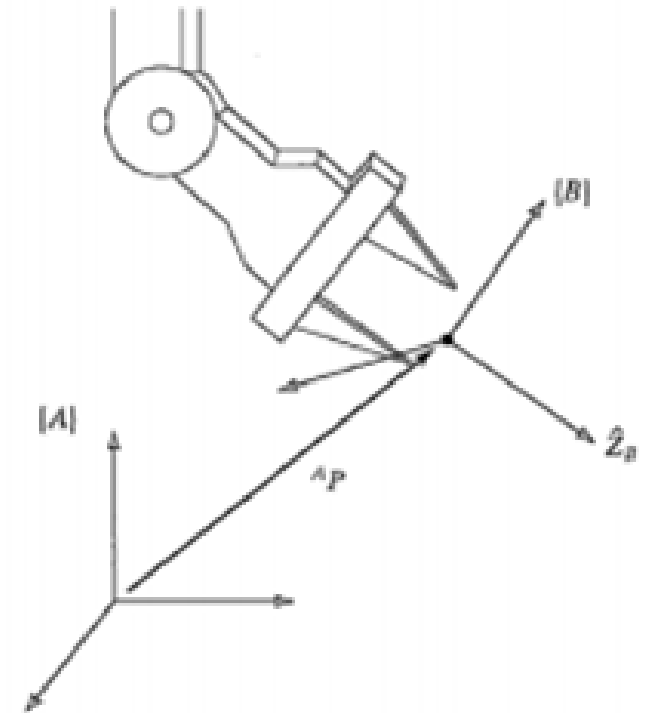
The orientation of the end-effector is described by the coordinate system $\{B\}$



Representing Position and Orientation

Description: Orientation

- The unit vectors of the coordinate system $\{B\}$ are $\hat{X}_B$, $\hat{Y}_B$, and $\hat{Z}_B$.
- If described in terms of the coordinate system $\{A\}$, they are ${}^A\hat{X}_B$, ${}^A\hat{Y}_B$ and ${}^A\hat{Z}_B$.

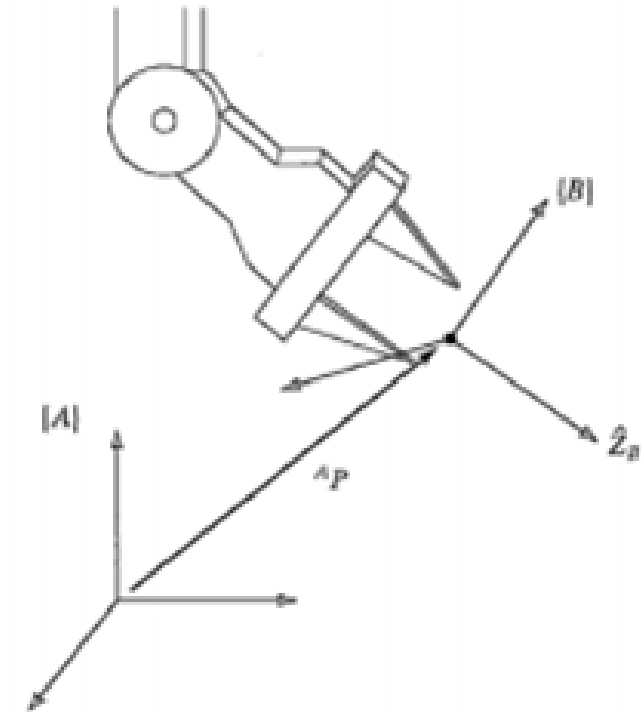




Representing Position and Orientation

Description: Orientation

- Rotation matrix:
It describes the orientation (or rotation) of a particular coordinate system with reference to another.
- The rotation matrix of any coordinate system contains its unit vectors described relative to the reference coordinate system.
- The unit vectors ${}^A\hat{X}_B$, ${}^A\hat{Y}_B$, and ${}^A\hat{Z}_B$ are written as the columns of the rotation matrix which describes $\{B\}$ relative to $\{A\}$.

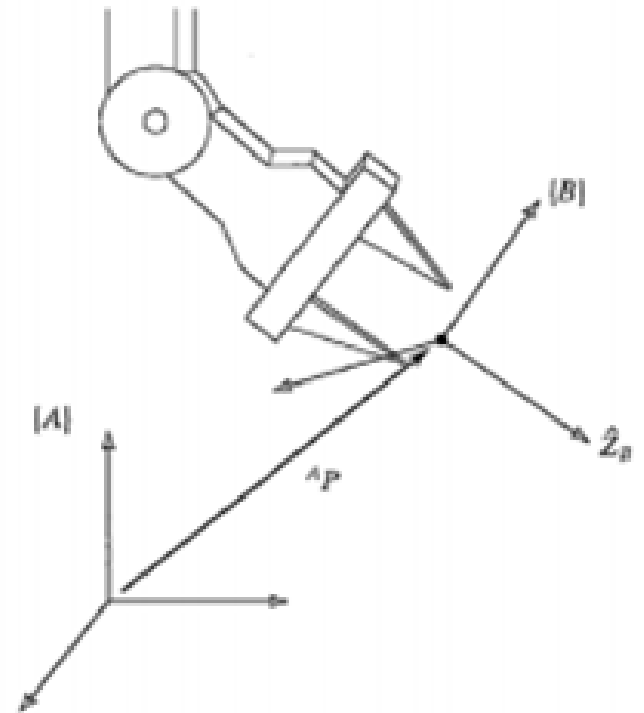


Representing Position and Orientation

Description: Orientation

- This rotation matrix which will describe $\{B\}$ relative to $\{A\}$ is called ${}^A_B R$.
- Since the rotation matrix of any coordinate system contains its unit vectors described relative to the reference coordinate system:

$${}^A_B R = \begin{bmatrix} \underline{{}^A\hat{X}_B} & \underline{{}^A\hat{Y}_B} & \underline{{}^A\hat{Z}_B} \end{bmatrix}$$





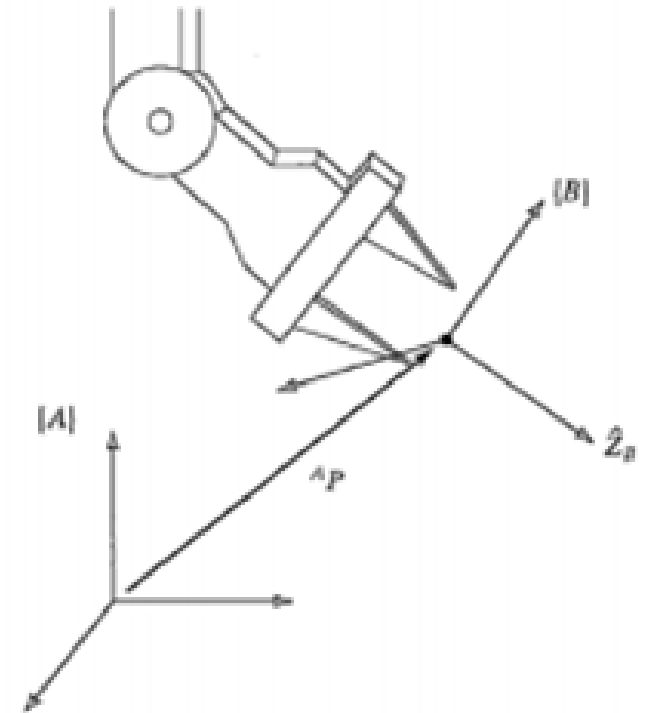
Representing Position and Orientation

Description: Orientation

$${}^A_B R = [{}^A\hat{X}_B \quad {}^A\hat{Y}_B \quad {}^A\hat{Z}_B]$$

Since the 3 unit vectors are the columns of a rotation matrix:

$${}^A_B R = [{}^A\hat{X}_B \quad {}^A\hat{Y}_B \quad {}^A\hat{Z}_B] = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$



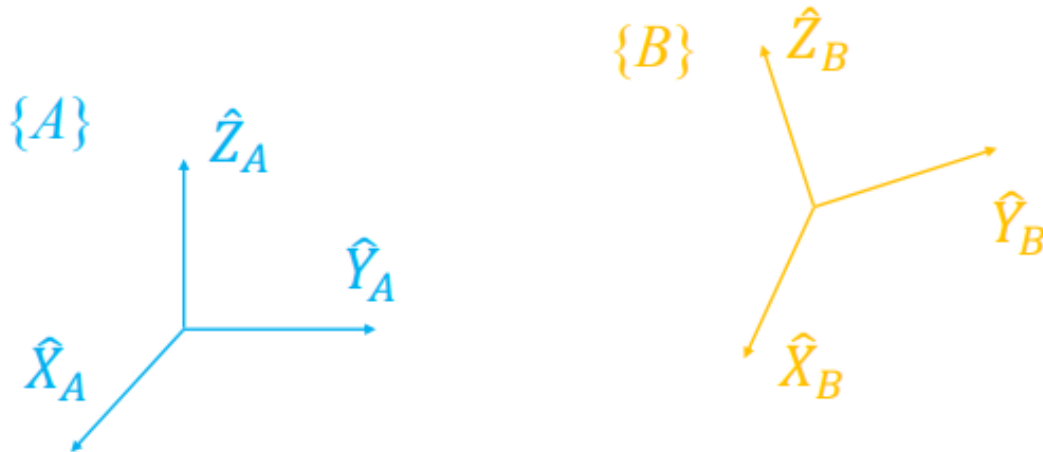


Representing Position and Orientation

Description: Orientation

$${}^A_B R = [{}^A\hat{X}_B \quad {}^A\hat{Y}_B \quad {}^A\hat{Z}_B] = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

We know that the components of any vector are its projections on the reference coordinate system's unit vectors.





Representing Position and Orientation

Description: Orientation

$${}^A_B R = [{}^A\hat{X}_B \quad {}^A\hat{Y}_B \quad {}^A\hat{Z}_B] = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

We know that the components of any vector are its projections on the reference coordinate system's unit vectors.

$${}^A\hat{X}_B = \begin{bmatrix} r_{11} \\ r_{21} \\ r_{31} \end{bmatrix} = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A \end{bmatrix} \quad {}^A\hat{Y}_B = \begin{bmatrix} r_{12} \\ r_{22} \\ r_{32} \end{bmatrix} = \begin{bmatrix} \hat{Y}_B \cdot \hat{X}_A \\ \hat{Y}_B \cdot \hat{Y}_A \\ \hat{Y}_B \cdot \hat{Z}_A \end{bmatrix}$$

$${}^A\hat{Z}_B = \begin{bmatrix} r_{13} \\ r_{23} \\ r_{33} \end{bmatrix} = \begin{bmatrix} \hat{Z}_B \cdot \hat{X}_A \\ \hat{Z}_B \cdot \hat{Y}_A \\ \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix}$$



Representing Position and Orientation

Description: Orientation

$${}^A_B R = [{}^A\hat{X}_B \quad {}^A\hat{Y}_B \quad {}^A\hat{Z}_B] = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix}$$

It can be seen that the first row is the unit vector $\hat{X}_A$ of the coordinate system $\{A\}$ expressed in $\{B\}$

The second row is the unit vector $\hat{Y}_A$ of the coordinate system $\{A\}$ expressed in $\{B\}$

The third row is the unit vector $\hat{Z}_A$ of the coordinate system $\{A\}$ expressed in $\{B\}$



Representing Position and Orientation

Description: Orientation

$${}^A_B R = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix}$$

Diagram illustrating the components of the rotation matrix ${}^A_B R$. The matrix is composed of dot products of unit vectors from frame B with unit vectors from frame A. The columns of the matrix correspond to the transformed axes of frame A, represented by the following expressions in circles:

- ${}^B \hat{X}_A^T$ (Top circle)
- ${}^B \hat{Y}_A^T$ (Middle circle)
- ${}^B \hat{Z}_A^T$ (Bottom circle)

Below the matrix, the rotation matrix is also shown as a column vector of these transformed axes:

$${}^A_B R = \begin{bmatrix} {}^B \hat{X}_A^T \\ {}^B \hat{Y}_A^T \\ {}^B \hat{Z}_A^T \end{bmatrix}$$



Representing Position and Orientation

Description: Orientation

$${}^A_B R = [{}^A\hat{X}_B \quad {}^A\hat{Y}_B \quad {}^A\hat{Z}_B] = \begin{bmatrix} {}^B\hat{X}_A^T \\ {}^B\hat{Y}_A^T \\ {}^B\hat{Z}_A^T \end{bmatrix}$$

$${}^B_A R = {}^A_B R^T$$

$${}^A_B R^T {}^A_B R = \begin{bmatrix} {}^A\hat{X}_B^T \\ {}^A\hat{Y}_B^T \\ {}^A\hat{Z}_B^T \end{bmatrix} [{}^A\hat{X}_B \quad {}^A\hat{Y}_B \quad {}^A\hat{Z}_B] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$$

$${}^A_B R = {}^B_A R^{-1} = {}^B_A R^T$$



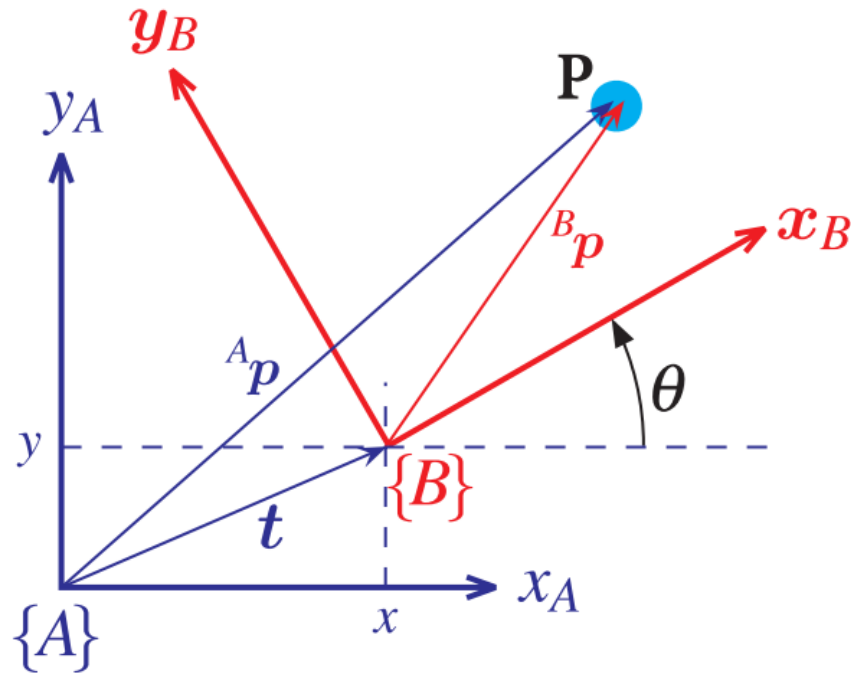
Representing Position and Orientation

- To summarize:
 - The position of any point is described using a vector
 - The orientation of any body is described using a matrix



Working in 2D

$$p = x\hat{x} + y\hat{y}$$



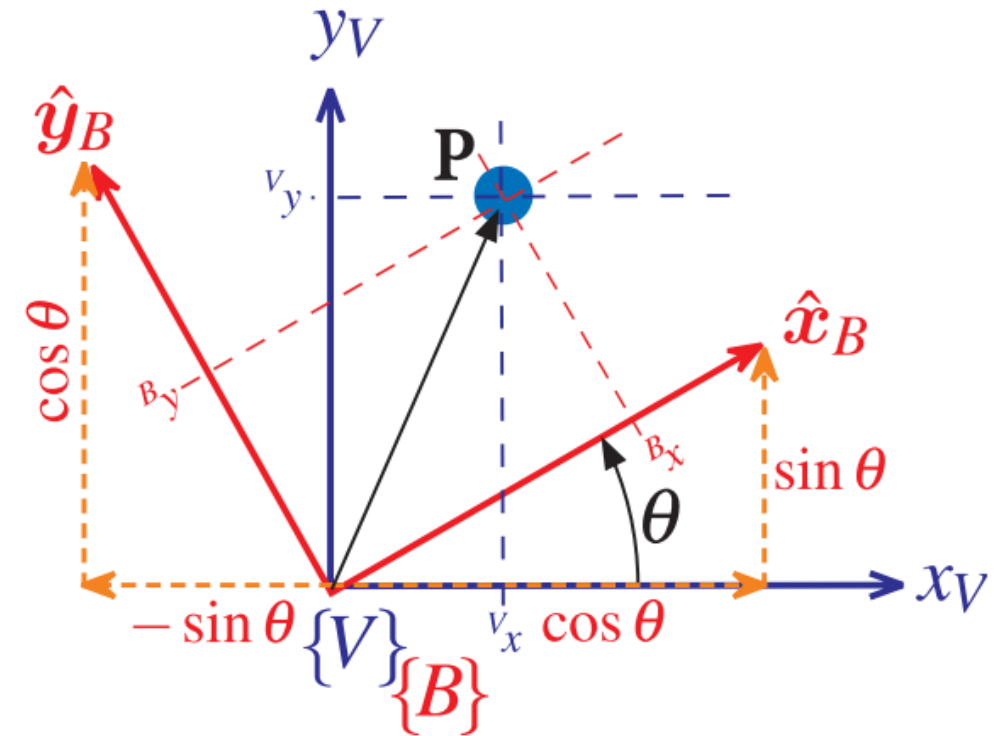
Describe frame {B} with respect to frame {A}



Working in 2D

We introduce another frame $\{V\}$ whose axis are parallel to frame $\{A\}$ but origin lies on frame $\{B\}$. Point $\mathbf{P}$ can be expressed in terms of frame $\{V\}$ as:

$$\begin{aligned} {}^V p &= {}^V x \hat{x}_V + {}^V y \hat{y}_V \\ &= (\hat{x}_V \ \hat{y}_V) \begin{pmatrix} {}^V x \\ {}^V y \end{pmatrix} \end{aligned} \quad \text{Eq 1}$$



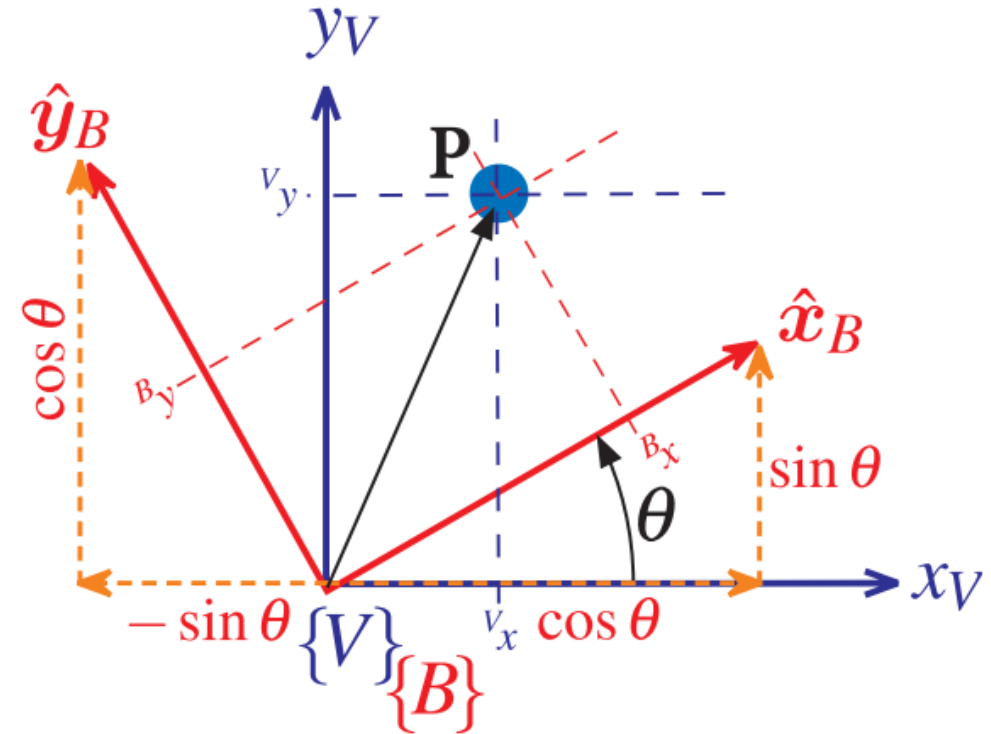
Working in 2D

Frame {B} can be described using its orthogonal axes as:

$$\begin{aligned}\hat{x}_B &= \cos\theta \hat{x}_V + \sin\theta \hat{y}_V \\ \hat{y}_B &= -\sin\theta \hat{x}_V + \cos\theta \hat{y}_V\end{aligned} = (\hat{x}_V \quad \hat{y}_V) \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

We can represent point **P** wrt frame {B} as:

$${}^B p = {}^B x \hat{x}_B + {}^B y \hat{y}_B = (\hat{x}_B \quad \hat{y}_B) \begin{pmatrix} {}^B x \\ {}^B y \end{pmatrix} = (\hat{x}_V \quad \hat{y}_V) \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} {}^B x \\ {}^B y \end{pmatrix} \quad \text{Eq 2}$$





Orthonormal Rotation Matrix

Equate equations 1 and 2, to see how points are transformed when frame is rotated.

$$\begin{pmatrix} {}^Vx \\ {}^Vy \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} {}^Bx \\ {}^By \end{pmatrix}$$



Rotation Matrix: transforms a point P from one frame to the other. It is denoted by ${}^V\mathbf{R}_B$

$$\begin{pmatrix} {}^Bx \\ {}^By \end{pmatrix} = ({}^V\mathbf{R}_B)^{-1} \begin{pmatrix} {}^Vx \\ {}^Vy \end{pmatrix} = ({}^V\mathbf{R}_B)^T \begin{pmatrix} {}^Vx \\ {}^Vy \end{pmatrix} = {}^B\mathbf{R}_V \begin{pmatrix} {}^Vx \\ {}^Vy \end{pmatrix}$$



Orthonormal Rotation Matrix

- Properties of rotation matrix

$${}^X R_Y(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

- it is *orthonormal* (also called *orthogonal*) since each of its columns is a unit vector and the columns are orthogonal. ▶
- the columns are the unit vectors that define the axes of the rotated frame Y with respect to X and are by definition both unit-length and orthogonal.
- it belongs to the special orthogonal group of dimension 2 or $R \in \mathbf{SO}(2) \subset \mathbb{R}^{2 \times 2}$. This means that the product of any two matrices belongs to the group, as does its inverse.
- its *determinant* is $+1$, which means that the length of a vector is unchanged after transformation, that is, $\|{}^Y \mathbf{p}\| = \|{}^X \mathbf{p}\|, \forall \theta$.
- the inverse is the same as the transpose, that is, $R^{-1} = R^T$.



Example

Figure 2.6 shows a frame $\{B\}$ that is rotated relative to frame $\{A\}$ about $\hat{Z}$ by 30 degrees. Here, $\hat{Z}$ is pointing out of the page.

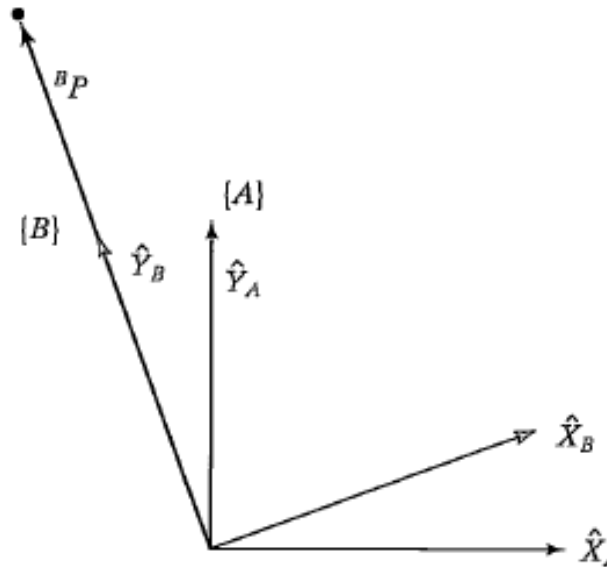


FIGURE 2.6: $\{B\}$ rotated 30 degrees about $\hat{Z}$.



Example

Writing the unit vectors of $\{B\}$ in terms of $\{A\}$ and stacking them as the columns of the rotation matrix, we obtain

$${}^A_B R = \begin{bmatrix} 0.866 & -0.500 & 0.000 \\ 0.500 & 0.866 & 0.000 \\ 0.000 & 0.000 & 1.000 \end{bmatrix}. \quad (2.14)$$

Given

$${}^B P = \begin{bmatrix} 0.0 \\ 2.0 \\ 0.0 \end{bmatrix}, \quad (2.15)$$

we calculate ${}^A P$ as

$${}^A P = {}^A_B R {}^B P = \begin{bmatrix} -1.000 \\ 1.732 \\ 0.000 \end{bmatrix}.$$



Three-Angle Representations

- Euler's rotation theorem states that *any* rotation can be represented by *not more than three* rotations about coordinate axes.
- This means that in general an arbitrary rotation between frames can be decomposed into a sequence of three rotation angles and associated rotation axes.
- Euler's rotation theorem requires successive rotation about three axes such that no two successive rotations are about the same axis.
- There are two classes of rotation sequence: Eulerian and Cardanian, named after Euler and Cardano respectively
 - Euler: XYX , XZX , YXY , YZY , ZXZ , or ZYZ
 - Cardanian: XYZ , XZY , YZX , YXZ , ZXY , or ZYX



Three-Angle Representations

- Euler-Angles: Most common sequence: ZYZ

$$R = R_z(\phi)R_y(\theta)R_z(\psi)$$

The Euler angles are the 3-vector $\boldsymbol{\Gamma} = (\phi, \theta, \psi)$.

- Cardanian Angles: roll-pitch-yaw sequence as ZYX or XYZ depending on if it is a mobile robot or a robotic arm.
- Convention for vehicles (ships, aircraft and cars): x-axis points in the forward direction and z-axis points either up or down.



Three-Angle Representations

- It is intuitive to apply the rotations in following sequence (ZYX angle sequence):
 1. yaw (direction of travel)
 2. pitch (elevation of the front with respect to horizontal) and
 3. roll (rotation about the forward axis of the vehicle)

$$\mathbf{R} = \mathbf{R}_z(\theta_y) \mathbf{R}_y(\theta_p) \mathbf{R}_x(\theta_r)$$

- Convention for robot gripper: the z-axis points forward and the x-axis is either up or down. This leads to the XYZ angle sequence

$$\mathbf{R} = \mathbf{R}_x(\theta_y) \mathbf{R}_y(\theta_p) \mathbf{R}_z(\theta_r)$$



Pose in 3D

- Similar to 2D case
- Transformation of frame {B} wrt frame {A}

$$\begin{pmatrix} {}^A x \\ {}^A y \\ {}^A z \\ 1 \end{pmatrix} = \begin{pmatrix} {}^A R_B & \mathbf{t} \\ \mathbf{0}_{1 \times 3} & 1 \end{pmatrix} \begin{pmatrix} {}^B x \\ {}^B y \\ {}^B z \\ 1 \end{pmatrix}$$

- If points are represented by homogeneous coordinate vectors

$$\begin{aligned} {}^A \tilde{\mathbf{p}} &= \begin{pmatrix} {}^A R_B & \mathbf{t} \\ \mathbf{0}_{1 \times 3} & 1 \end{pmatrix} {}^B \tilde{\mathbf{p}} \\ &= {}^A T_B {}^B \tilde{\mathbf{p}} \end{aligned}$$

${}^A T_B$ is a 4×4 homogeneous transformation matrix



Translation, Rotation and Transformation

- Translation is basically moving a point in space along a finite distance along a given direction
- Rotation is basically a rotational operator which rotates a vector and turns it into a new vector
- A rotation matrix can be interpreted as a matrix which rotates a vector through a set of angles along the axes
- Or, it can be seen as a rotation matrix which describes the orientation of a frame relative to a reference frame



Translation, Rotation and Transformation

- Suppose we have a vector ${}^A P_1$ described relative to a frame $\{A\}$
- This vector is now to be rotated so it is called ${}^A P_2$

$${}^A P_2 = R {}^A P_1$$

Another form for this expression is:

$${}^A P_2 = R_k(\theta) {}^A P_1$$

$R_k(\theta)$ is the rotational operator which performs rotation about an axis $\hat{K}$ by θ degrees.

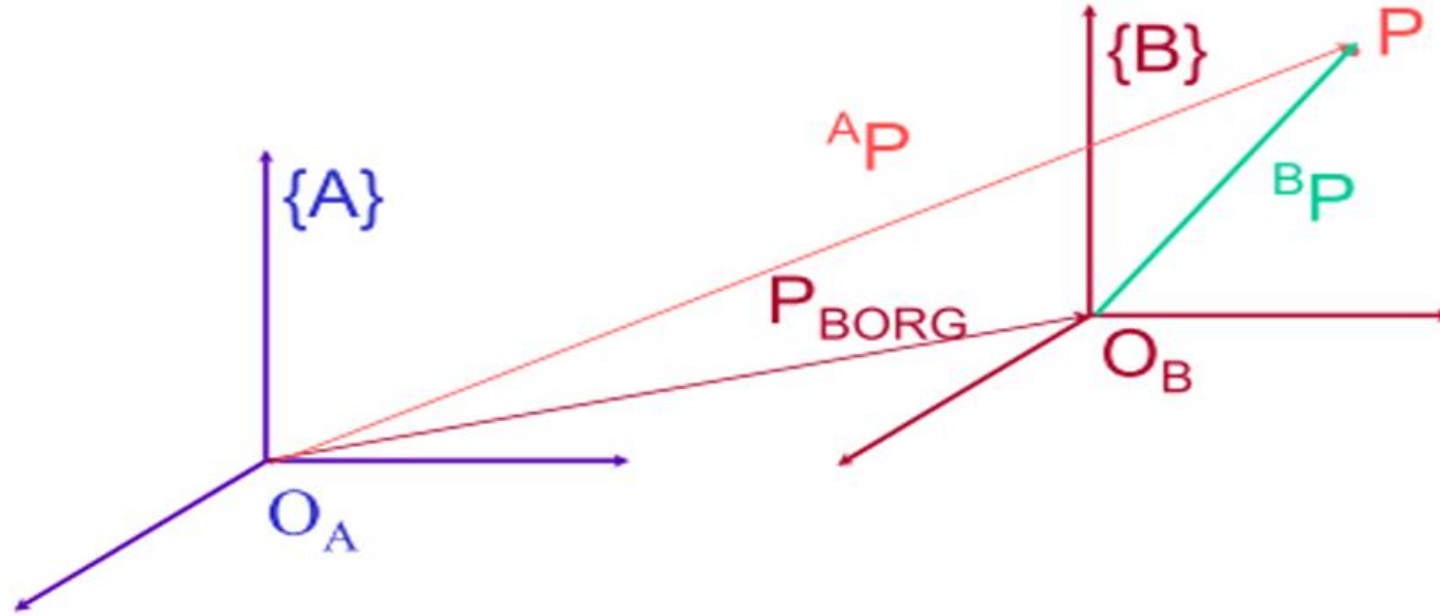


Translation, Rotation and Transformation

This rotation matrix may be written as the transformation matrix whose position vector part is zero since there is no translation.

$$\begin{bmatrix} {}^A P_2 \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A_B R & \cancel{{}^A P_{TR}} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^A P_1 \\ 1 \end{bmatrix}$$

Translations



changing the position description of a point P

$$\overrightarrow{O_B P} \longrightarrow \overrightarrow{O_A P}$$

(Two different vectors)

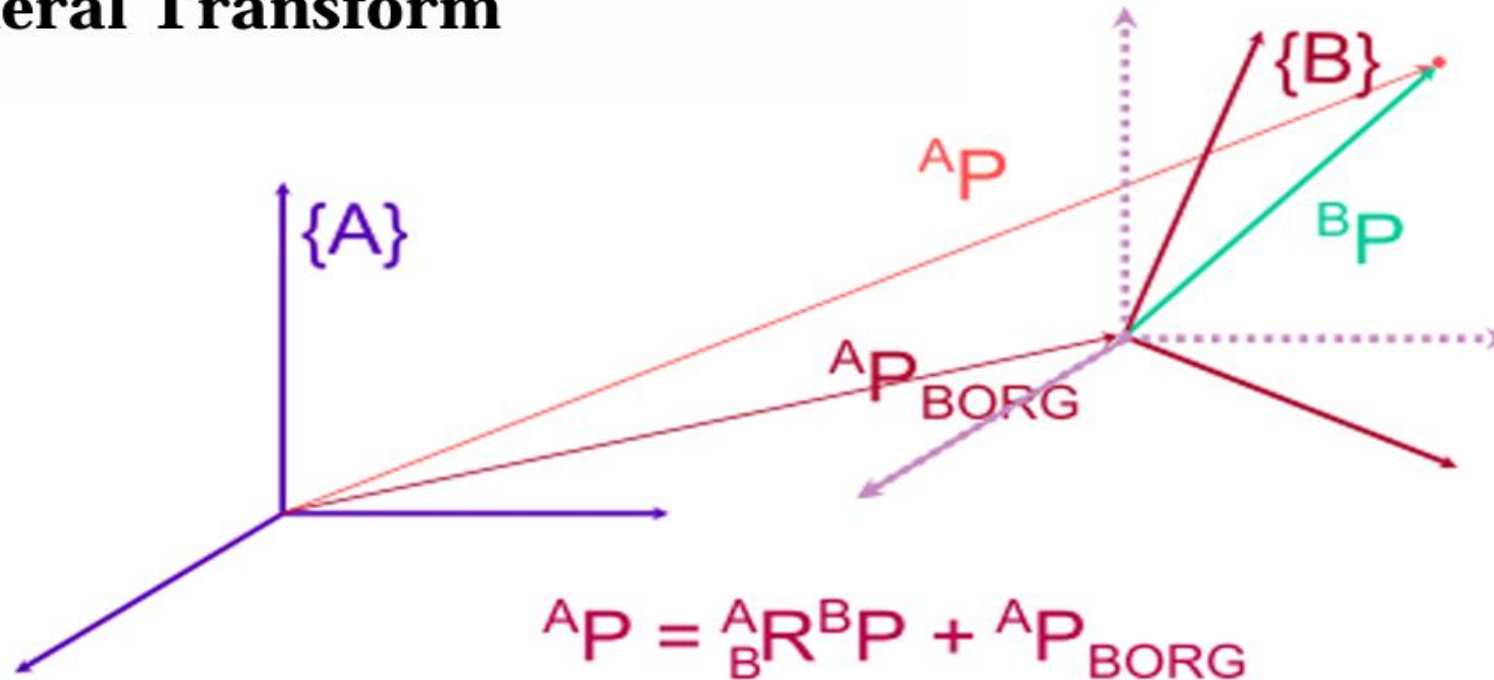
$$P_{BORG} : P_{O_B} \longrightarrow P_{O_A}$$

$$P_{O_A} = P_{O_B} + P_{BORG}$$



Translations

- **General Transform**



$${}^A P = {}^A_B R {}^B P + {}^A P_{BORG}$$
$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A_B R & | & {}^A P_{BORG} \\ \hline 0 & 0 & 0 & | & 1 \end{bmatrix} \begin{bmatrix} {}^B P \\ 1 \end{bmatrix}$$

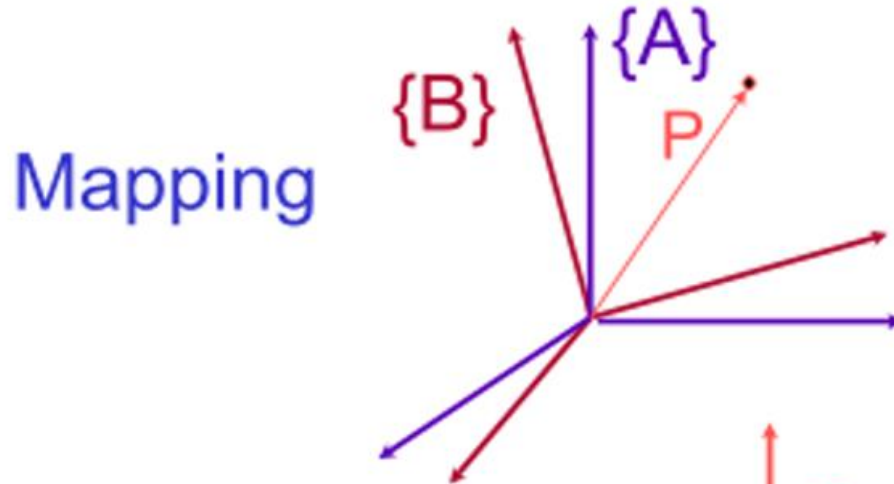
$$\underline{\underline{{}^A P = {}^A_B T {}^B P}}$$

$\begin{matrix} (4 \times 1) & (4 \times 4) & (4 \times 1) \end{matrix}$



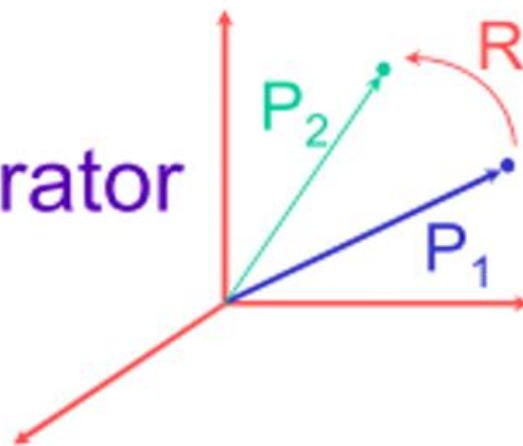
Translations

- **Mapping:** changing descriptions from frame to frame
- **Operators:** moving points (within the same frame)



$${}^A P = {}^A_B R {}^B P$$

Rotational Operator



$$R: P_1 \longrightarrow P_2$$

$$P_2 = R P_1$$



Translations

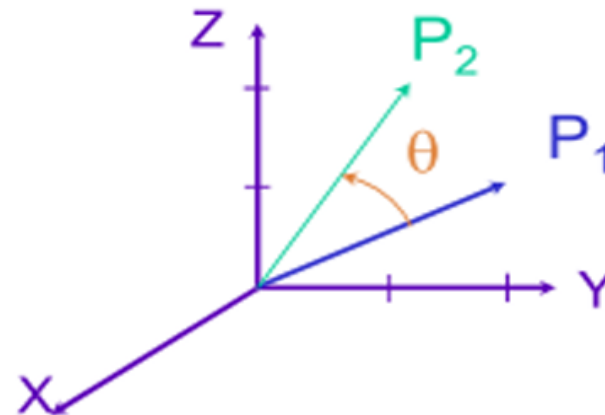
- Rotational Operators**

$$R_K(\theta): P_1 \longrightarrow P_2$$

$$P_2 = R_K(\theta) P_1$$

Example

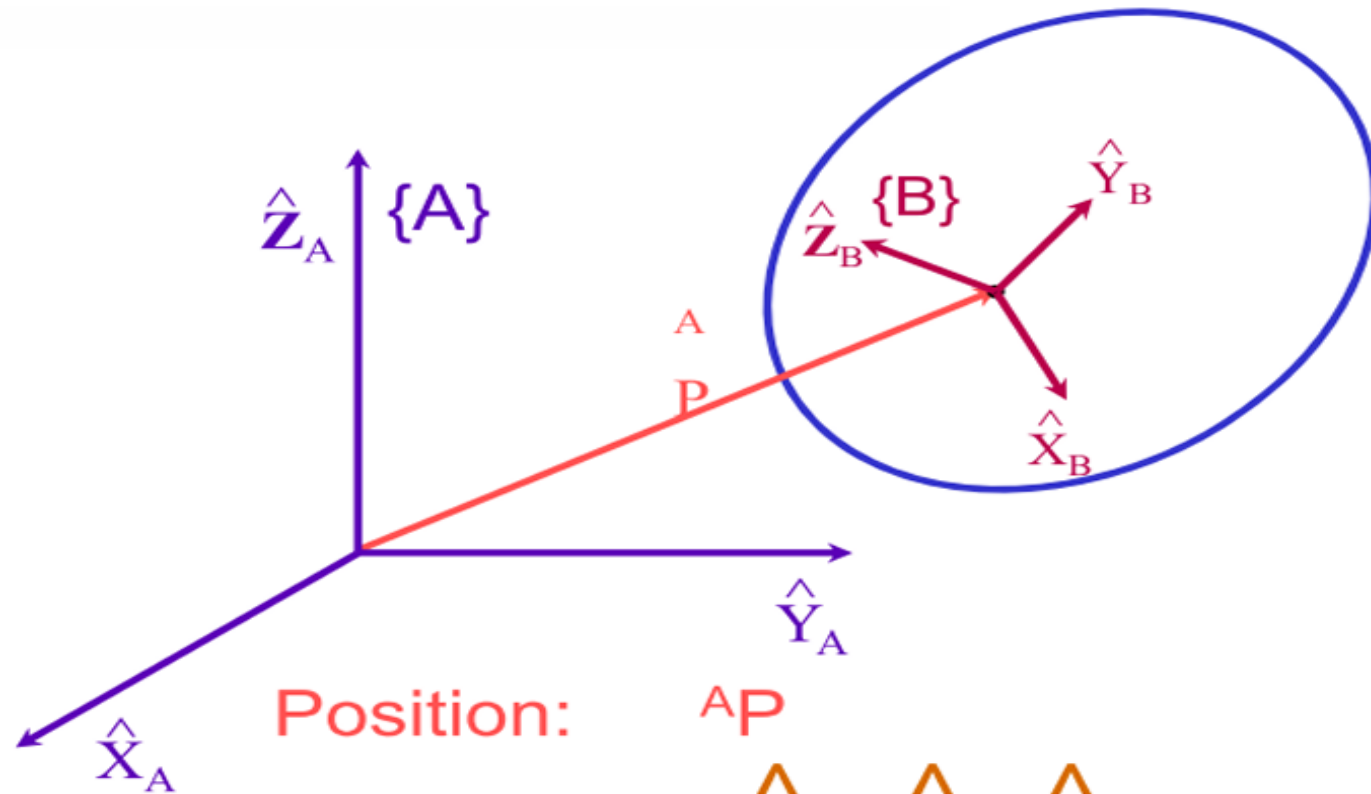
$$R_X(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$



$$P_2 = R_X(\theta)P_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.8 & -0.6 \\ 0 & 0.6 & 0.8 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$



Rotation Matrix



Position: ${}^A P$

Orientation: $\{{}^A \hat{X}_B, {}^A \hat{Y}_B, {}^A \hat{Z}_B\}$

describes rotations of $\{B\}$ with respect to $\{A\}$

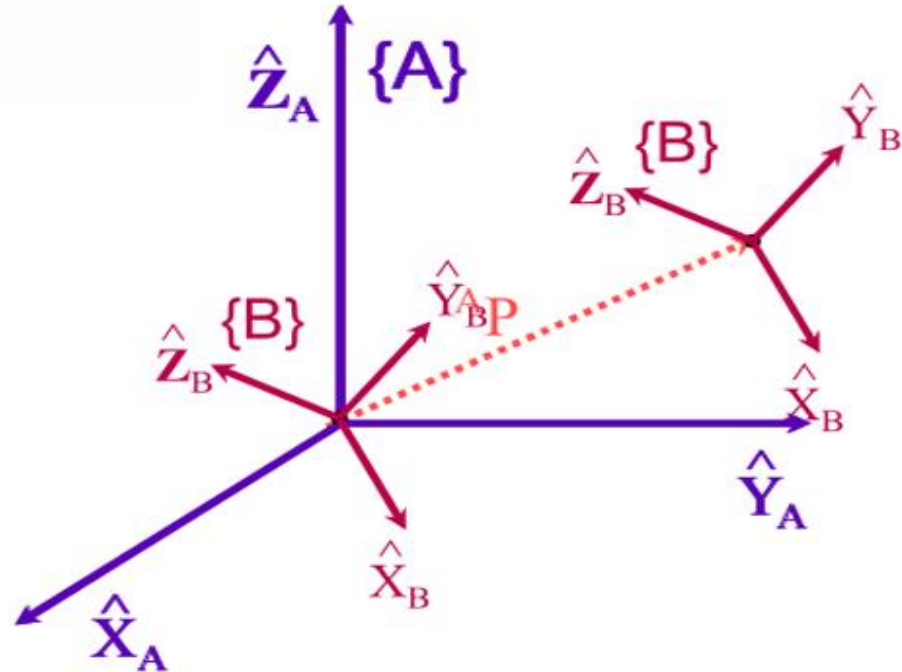


Rotation Matrix



Rotation Matrix

$${}^A_B R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$



$${}^A \hat{X}_B = {}^A_B R \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad {}^A \hat{Y}_B = {}^A_B R \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad {}^A \hat{Z}_B = {}^A_B R \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \longrightarrow \quad {}^A_B R = \begin{bmatrix} {}^A \hat{X}_B & {}^A \hat{Y}_B & {}^A \hat{Z}_B \end{bmatrix}$$

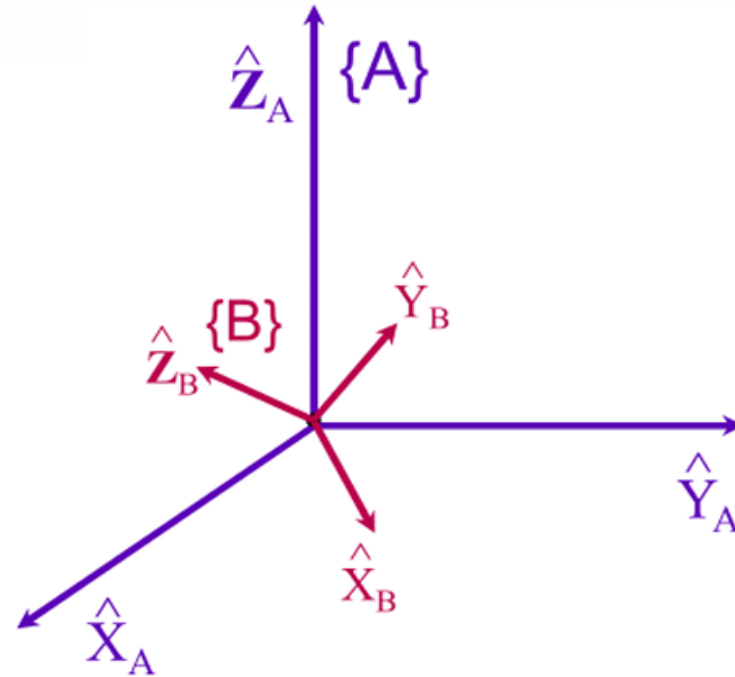


Rotation Matrix

$${}^A_B R = \begin{bmatrix} {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \end{bmatrix}$$

Dot Product

$${}^A\hat{X}_B = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A \end{bmatrix}$$

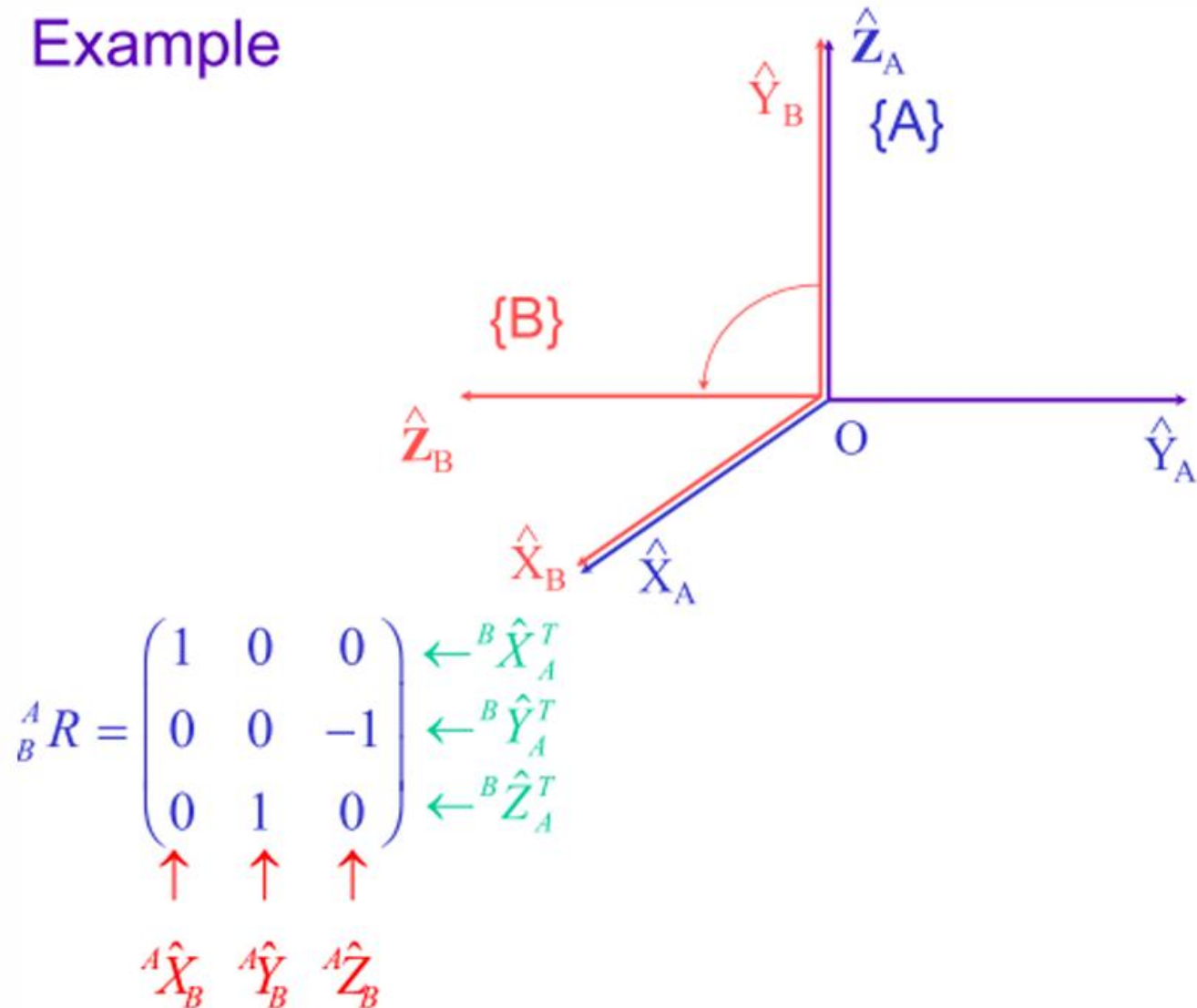


$${}^A_B R = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} \quad {}^B X_A^T$$



Rotation Matrix

Example





Rotation Matrix

$${}^A_B R = \begin{bmatrix} {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \end{bmatrix} = \begin{bmatrix} {}^B\hat{X}_A^T \\ {}^B\hat{Y}_A^T \\ {}^B\hat{Z}_A^T \end{bmatrix} = \begin{bmatrix} {}^B\hat{X}_A & {}^B\hat{Y}_A & {}^B\hat{Z}_A \end{bmatrix}^T = {}^B_A R^T$$

$$\underline{\underline{{}^A_B R = {}^B_A R^T}}$$

Inverse of Rotation Matrices

$${}^A_B R^{-1} = {}^B_A R = {}^A_B R^T$$

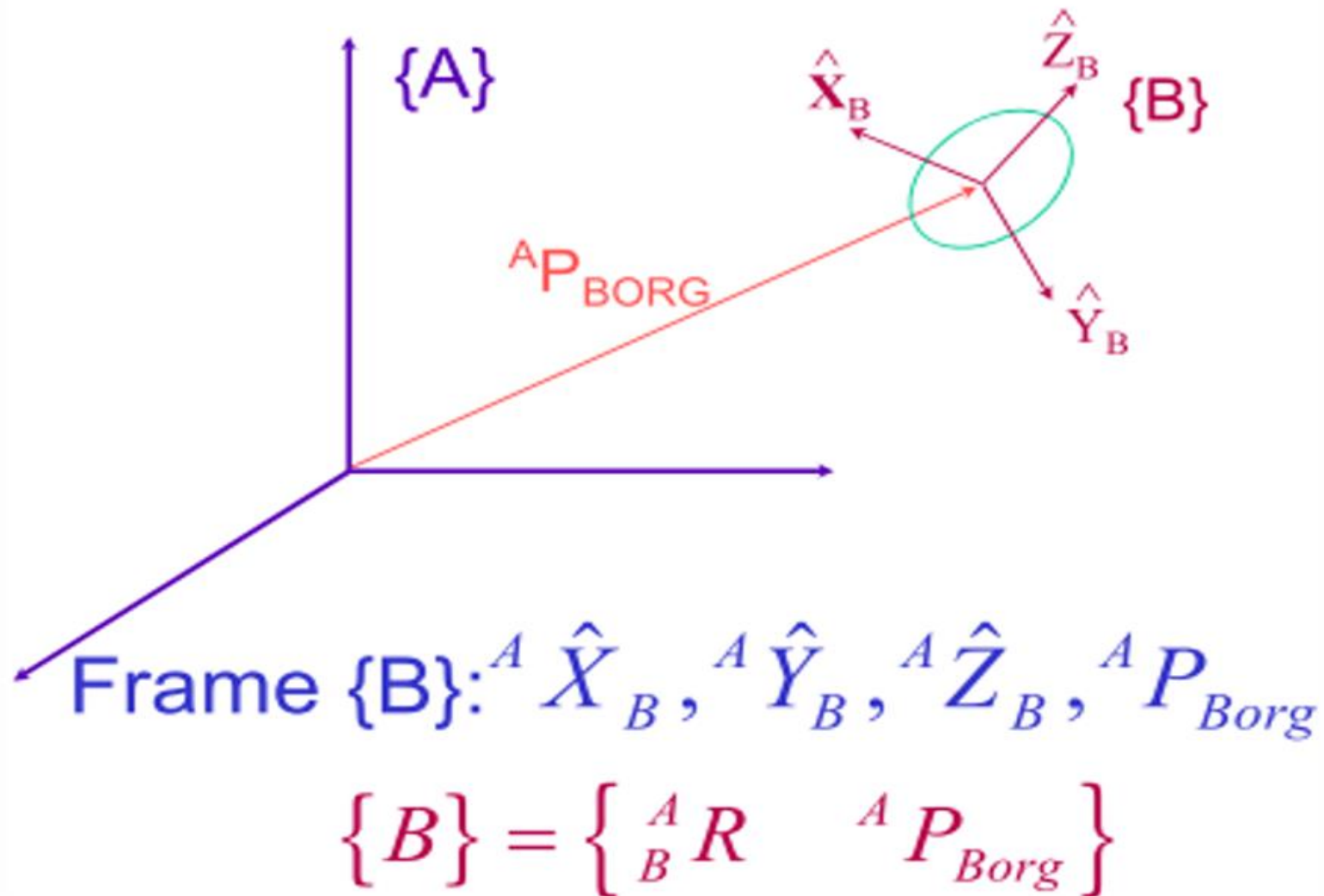
$$\boxed{{}^A_B R^{-1} = {}^A_B R^T}$$

Orthonormal Matrix



Rotation Matrix

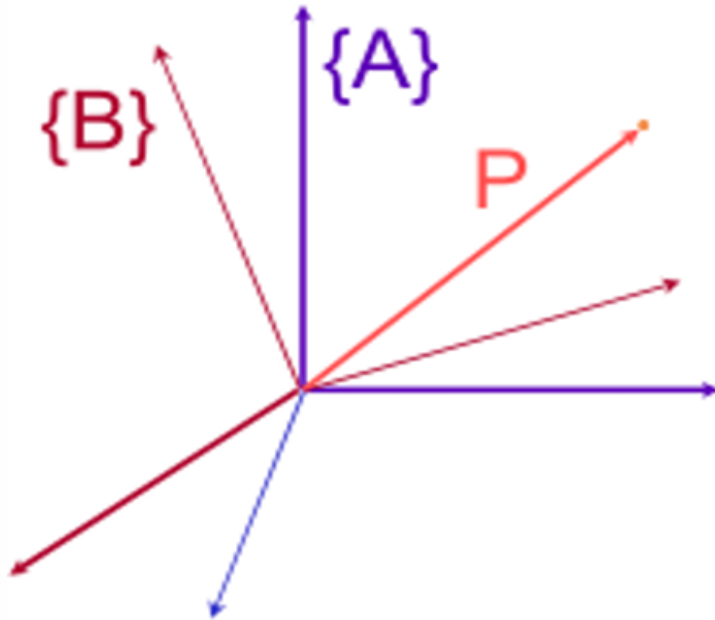
- Description of a Frame with respect to another frame





Rotation Matrix

- Mapping changing descriptions from frame to frame
- Rotations



If P is given in $\{B\}$: ${}^B P$

$${}^A P = \begin{pmatrix} {}^B \hat{X}_A \cdot {}^B P \\ {}^B \hat{Y}_A \cdot {}^B P \\ {}^B \hat{Z}_A \cdot {}^B P \end{pmatrix} = \begin{pmatrix} {}^B \hat{X}_A^T \\ {}^B \hat{Y}_A^T \\ {}^B \hat{Z}_A^T \end{pmatrix} \cdot {}^B P$$



$${}^A P = {}^A_B R {}^B P$$



Rotation Matrix

Orthonormal rotation matrices along each axis:

$$R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}$$

$$R_y(\theta) = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$$

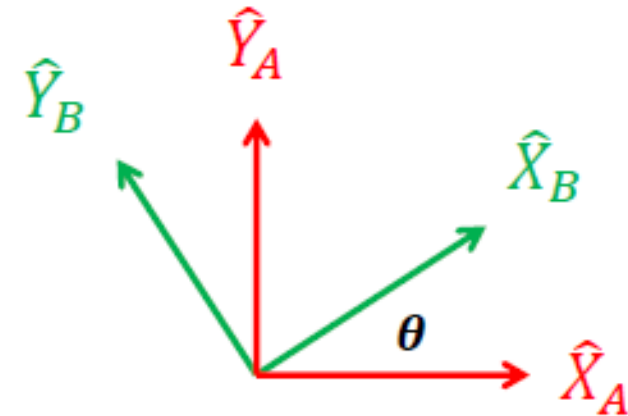
$$R_z(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



Translation, Rotation and Transformation

For example, if we require mapping from one frame to another which differ only by a rotation about the z-axis:

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$





Translation, Rotation and Transformation

The homogenous transform can be interpreted as the transformation operator which *rotates* and *translates* a vector ${}^A P_1$ to compute a new vector ${}^A P_2$:

$${}^A P_2 = T {}^A P_1$$

It can also be thought of as a transform that describes:

1. How a frame rotates through a set of angles
2. How the frame is translated by a certain offset from a reference frame.

Example 1

Figure 2.8 shows a frame $\{B\}$, which is rotated relative to frame $\{A\}$ about $\hat{Z}$ by 30 degrees, translated 10 units in $\hat{X}_A$, and translated 5 units in $\hat{Y}_A$. Find ${}^A P$, where ${}^B P = [3.07.00.0]^T$.

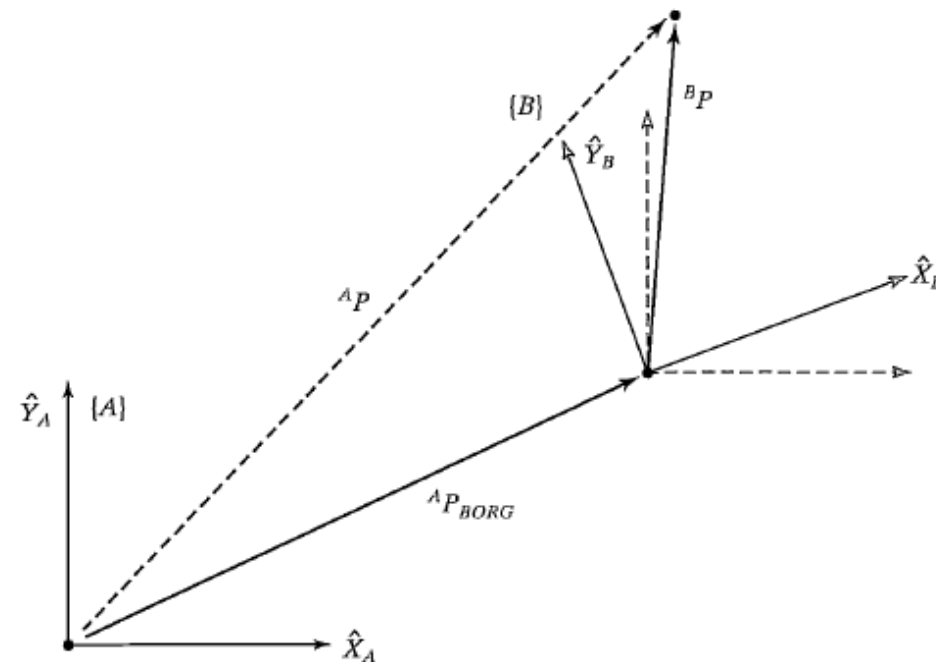


FIGURE 2.8: Frame $\{B\}$ rotated and translated.



Example 1

The definition of frame $\{B\}$ is

$${}^A_B T = \begin{bmatrix} 0.866 & -0.500 & 0.000 & 10.0 \\ 0.500 & 0.866 & 0.000 & 5.0 \\ 0.000 & 0.000 & 1.000 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Given

$${}^B P = \begin{bmatrix} 3.0 \\ 7.0 \\ 0.0 \end{bmatrix},$$

we use the definition of $\{B\}$ just given as a transformation:

$${}^A P = {}^A_B T {}^B P = \begin{bmatrix} 9.098 \\ 12.562 \\ 0.000 \end{bmatrix}.$$

Example 2

Figure 2.10 shows a vector ${}^A P_1$. We wish to compute the vector obtained by rotating this vector about $\hat{Z}$ by 30 degrees. Call the new vector ${}^A P_2$.

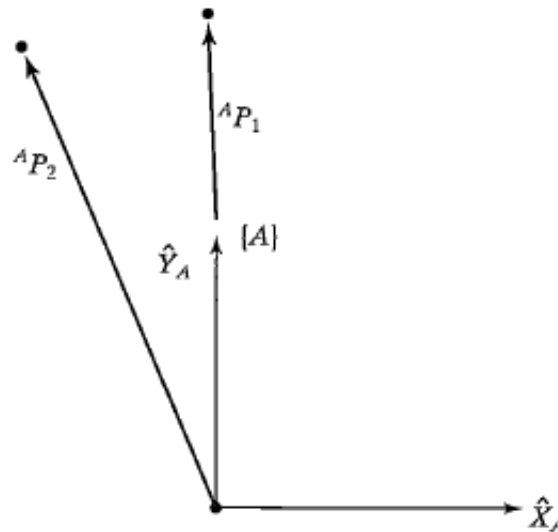


FIGURE 2.10: The vector ${}^A P_1$ rotated 30 degrees about $\hat{Z}$.



Example 2

The rotation matrix that rotates vectors by 30 degrees about $\hat{Z}$ is the same as the rotation matrix that describes a frame rotated 30 degrees about $\hat{Z}$ relative to the reference frame. Thus, the correct rotational operator is

$$R_z(30.0) = \begin{bmatrix} 0.866 & -0.500 & 0.000 \\ 0.500 & 0.866 & 0.000 \\ 0.000 & 0.000 & 1.000 \end{bmatrix}. \quad (2.30)$$

Given

$${}^A P_1 = \begin{bmatrix} 0.0 \\ 2.0 \\ 0.0 \end{bmatrix}, \quad (2.31)$$

we calculate ${}^A P_2$ as

$${}^A P_2 = R_z(30.0) {}^A P_1 = \begin{bmatrix} -1.000 \\ 1.732 \\ 0.000 \end{bmatrix}$$



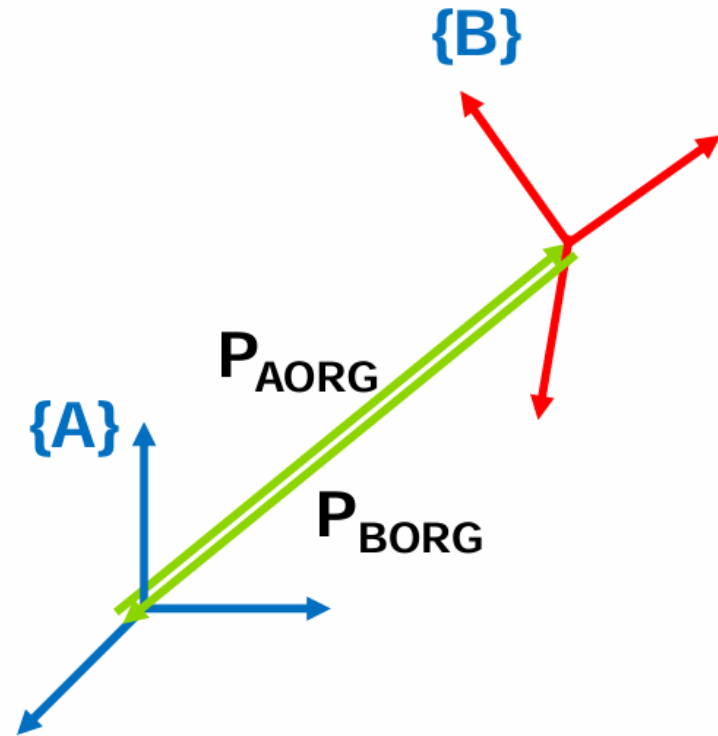
Transformation

- Inverse Transform:**

$${}^A_B T = \begin{bmatrix} {}^A_B R & {}^A P_{Borg} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R^{-1} = R^T \quad T^{-1} \neq T^T$$

$${}^A_B T^{-1} = {}^B_A T = \begin{bmatrix} {}^A_B R^T & -{}^A_B R^T {}^A P_{Borg} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^B P_{Aorg}$$



Any Questions?

